

Proof of 2nd part of Gersgorin's theorem

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Statement. Let $A = [a_{ij}]$ be any $n \times n$ matrix. For each i , let,

$$r_i = \sum_{\substack{j=1 \\ i \neq j}}^n |a_{ij}|, \quad D_i = \{z : |z - a_{ii}| \leq r_i\}, \text{ then,}$$

1. All the eigen values of A lie in the region $\mathcal{D} = \bigcup_{i=1}^n D_i$
2. If a connected component of $\mathcal{D}$ is made of k discs, then it contains exactly k eigen values of A .

Proof (2) For the given matrix $A = [a_{ij}]$ choose the following matrix valued function $A(t)$ on $(0, 1)$ with $A(1) = A$.

$$A_{ij}(t) = \begin{cases} a_{ii}, & i = j \\ t \cdot a_{ij}, & i \neq j \end{cases}$$

Consequently, the eigenvalues of the matrix

$A(t)$ form a continuous function on $(0, 1)$, namely, $e_1(t), \dots, e_n(t)$. And $A(0)$ is the diagonal matrix $d(a_{11}, \dots, a_{nn})$.

Also, consider the usual metric on $\mathbb{C}$ such that for $x, y \in \mathbb{C}$, and $B \subset \mathbb{C}$, $\|x - y\|$ is the usual distance, and $\|x - B\| := \inf\{d(x, b) \mid b \in B\}$.

Now, apply *Gersgorin* to the matrix $A(t)$ to obtain n discs $D_i(t)$, for $1 \leq i \leq n$, centered at a_{ii} with radius $r_i(t)$ given by,

$$r_i(t) = \sum_{\substack{j=1 \\ j \neq i}}^n |A_{ij}(t)| = t \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}|$$

Observe $r_i(t)$ is increasing in t . And all i -discs are centered at a_{ii} . Hence, the discs $D_i(0)$ corresponding to the diagonal matrix $A(0)$ are the singletons $\{a_{ii}\}$, which gradually expand in size until they form the discs $D_i(1)$ corresponding to the original matrix $A(1) = A$. In particular for a union D of such discs, if $t < t'$, we have $D(t) \subset D(t')$.

Now given the connected component D made of k discs, let i be such that the disc centered at a_{ii} is a part of D . In our construction, recall, $D(t)$ is the union of k smaller discs $D_j(t)$ corresponding to the matrix $A(t)$. Say $e_i(1) \notin D(1) = D$. Then $e_i(1) \in D'$, for some other connected component D' . But as D and D' are disjoint, there exists $M > 0$ for which, $\|D - D'\| > M$. In particular, $\|e_i(1) - D(1)\| > M$.

Now the set $S = \{t : e_i(t) \in D(t)\}$ is non-empty as $e_i(0) = a_{ii} \in D(0)$, or $0 \in S$. So we can obtain $T = \sup S$. Note that T is as such that $e_i(T)$ is arbitrarily close to $D(T)$ (*formal proof at the end). Now, as $T < 1$, $D(T) \subset D(1)$. But, $e_i(1)$ is not arbitrarily close to $D(1)$, as $\|e_i(1) - D(1)\| > M$. Hence, $D(1) \neq D(T)$, or $D(T) \subsetneq D(1)$, or $D(T)$ is “strictly” inside $D(1)$.

But again, as $e_i(T)$ is arbitrarily close to $D(T)$, $e_i(T)$ is also “strictly” inside $D(1)$. Hence by applying the intermediate value theorem to e_i , we can find $t \in (T, 1)$ such that,

$$e_i(t) \in D(1), \text{ but, } e_i(t) \notin D(t), \text{ as, } t > T$$

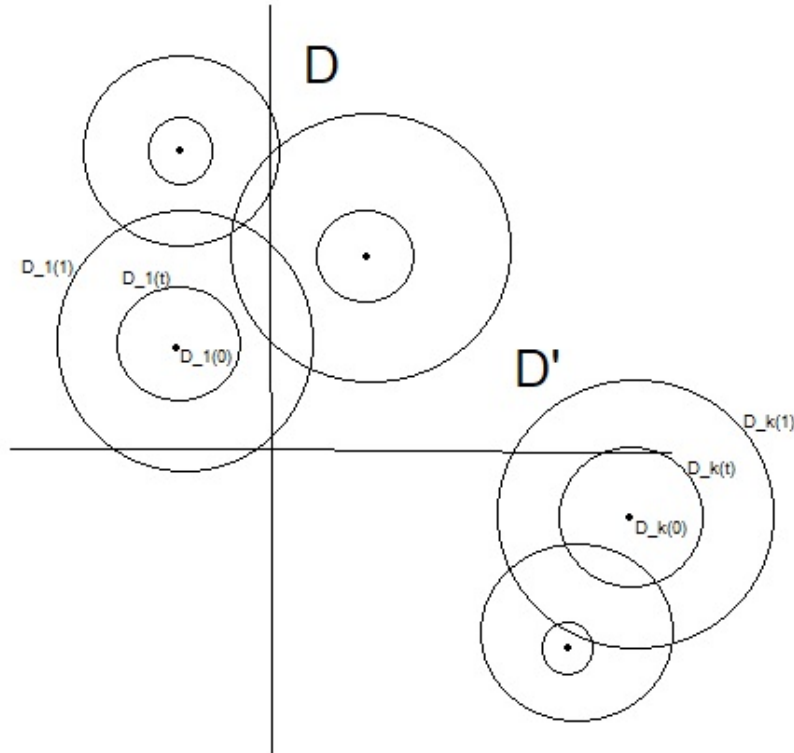
But $e_i(t)$ is an eigen value of $A(t)$. Hence by *Gersgorin* it must lie in some union of discs $D''(t) \neq D(t)$. But by our construction $D''(t)$ must lie in some “other” connected component D'' corresponding to the matrix A , since all intermediate discs which could grow up to D were parts of $D(t)$. Therefore, we have,

$$e_i(t) \in D, \text{ or, } e_i(t) \in D'', \text{ or, } e_i(t) \in D \cap D''$$

However, by definition of connected components $D \cap D'' = \emptyset$. Contradiction!

Or $e_i(1) \in D$. This is true for all k i 's for which the disc centered at a_{ii} was a part of D . Hence D contains at least k eigen values of A . However observe that the union of all the other connected components D' are together made of $n - k$ discs, and must have at least $n - k$ eigen values among themselves. Since there are only n eigen values possible for A we can conclude that D contains exactly k eigen values of A . $\square$

Image for reference.



* If $S = \{t : e_i(t) \in D(t)\}$, and $T = \sup S$, then $e_i(T)$ is arbitrarily close to $D(T)$.

Proof. Say this is not true. Then there exists $M > 0$ for which $\|e_i(T) - D(T)\| > M$, or $e_i(T)$ is strictly outside $D(T)$. By intermediate value theorem again, there exists ϵ such that $e_i(t)$ is still outside $D(T)$ for $t \in (T - \epsilon, T]$. But $D(t) \subset D(T)$ for every $t < T$. Therefore, $e_i(t) \notin D(t)$ for $t \in (T - \epsilon, T]$, or $\sup S < T - \epsilon$, which is bad! The claim therefore holds. $\square$